

DS02A - CORRECTION

Exercice 1

```
def f(a,b):
    if a < b:
        return a
    else :
        return b
assert f(3,7) == 3
```

Exercice 2

```
def f(x):
    for i in range(6):
        x = x + i
    return x
assert f(1) == 16
```

Exercice 3

```
mon_tuple = (1 ,6 ,8 ,3 ,2)
def f(un_tuple : tuple ):
    a = 0
    for elt in un_tuple :
        a = a + elt
    return a
assert f(mon_tuple) == 20
```

Exercice 4

```
def f(x):
    i = 0
    a = 0
    while i < x:
        a = a + 3
        i = i + 1
    return a
assert f(3) == 9
```

Exercice 5

```
def f(n):
    for i in range(n+1):
        if i % 2 == 0:
            print(i , end = " ")
f(9)      # 0 2 4 6 8
```

Exercice 6

```
def recherche_maximum_I(une_liste):
    n = len(une_liste)
    maximum = une_liste[0]
    for i in range(1,n):
        if une_liste[i] > maximum:
            maximum = une_liste[i]
    return maximum
ma_liste = [3 ,6 ,8 ,1 ,9 ,4 ,7]
assert recherche_maximum_I(ma_liste) == 9
```

```
def recherche_maximum_E(une_liste):
    maximum = une_liste[0]
    for elt in une_liste:
        if elt > maximum:
            maximum = elt
    return maximum
ma_liste = [3 ,6 ,8 ,1 ,9 ,4 ,7]
assert recherche_maximum_E(ma_liste) == 9
```

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DS02A - CORRECTION

Exercice 1

```
def f(a,b):
    if a < b:
        return a
    else :
        return b
assert f(7,3) == 3
```

Exercice 2

```
def f(x):
    for i in range(6):
        x = x + i
    return x
assert f(2) == 17
```

Exercice 3

```
mon_tuple = (1 ,6 ,7 ,3 ,2)
def f(un_tuple : tuple ):
    a = 0
    for elt in un_tuple :
        a = a + elt
    return a
assert f(mon_tuple) == 19
```

Exercice 4

```
def f(x):
    i = 0
    a = 0
    while i < x:
        a = a + 2
        i = i + 1
    return a
assert f(3) == 6
```

Exercice 5

```
def f(n):  
    for i in range(n+1):  
        if i % 2 == 0:  
            print(i , end = " ")  
f(7)    # 0 2 4 6
```

Exercice 6

```
def recherche_minimum_I(une_liste):  
    n = len(une_liste)  
    minimum = une_liste[0]  
    for i in range(1,n):  
        if une_liste[i] < minimum:  
            minimum = une_liste[i]  
    return minimum  
ma_liste = [3 ,6 ,8 ,1 ,9 ,4 ,7]  
assert recherche_minimum_I(ma_liste) == 1  
  
def recherche_minimum_E(une_liste):  
    minimum = une_liste[0]  
    for elt in une_liste:  
        if elt < minimum:  
            minimum = elt  
    return minimum  
ma_liste = [3 ,6 ,8 ,1 ,9 ,4 ,7]  
assert recherche_minimum_E(ma_liste) == 1
```